An explicit strong-coupling gap for the Wilson transfer matrix: $SU(3)$ is gapped for $g^2 \geq 1059$, with $\Delta \geq (\hbar c/a)\, 4\log[1/(176(e^{6/g^2}-1))]$

Correction (2026-10-02). A referee report and the corpus audit §3c found the errors below; the text is corrected in place.

- The former headline, $g^2 \geq 176$ with $\Delta_W \geq (\hbar c/a)\, 4\log(g^2/176)$, charged each plaquette its leading fundamental coefficient $1/g^2$, and no argument here bounds every polymer weight by that activity. It is kept only as the leading-activity figure. The rigorous threshold of the same chain is $g^2 \geq 1059$ (a sufficient integer bound); the former rigorous figure $1056 = 6 \times 176$ linearized $e^{6/g^2} - 1 \simeq 6/g^2$.
- §1 bounded $\sum_{r\neq 0} d_r c_r/c_0$, a sum of Fourier coefficients. Polymer weights need the sup norm of $f/c_0 - 1$, which by positivity of the c_r equals $\sum_{r\neq 0} d_r^2 c_r/c_0 = e^{\beta_W}/c_0 - 1$. The bound $e^{\beta_W} - 1$ survives with the new derivation.
- §1 called $d_f c_f/c_0 \simeq 1/g^2$ the tube activity. Its own Euler-characteristic formula gives $u = c_f/c_0 \simeq 1/(3g^2) = \beta_W/18$. The factor 6 between the two readings of the old table is $2d_f$ (the fundamental and its conjugate, and $|\chi_f| \leq d_f$); the former account of it is withdrawn. The $SU(N)$ formula holds for $N \geq 3$.
- The cluster size $4T$ was inferred from a single tube. §3 now proves it for chains of polymers and for polymers carrying the observables, on spatial tori with sides of at least 4. The plaquette-animal bound and the minimal polymer size 6 now carry proofs (§2), and the decay weight is written $d(\gamma) = \delta|\gamma|$ where the text had $a(\gamma) = (1+\delta)|\gamma|$.
- Withdrawn as unsupported: "$20e$ is the only constant with room" and "a careful count would lower 176 to about 10^2". The weight $a(\gamma)$ and the sup norm also have room (§3).
- §5 placed the strong edge of the intermediate region at the heuristic 176; it now uses the rigorous figures. §4's anisotropic transfer matrix now reads $e^{-a_t H_{\mathrm{KS}}/\hbar c}$.

Later-note pointers. The Dobrushin note proves the lower sufficient threshold $g^2 > 444$ for the same operator, with exact rate $\log[1/(18(e^{24/g^2}-1))]$ and leading estimate $\log(g^2/432)$. Openings §2, Opening 1, reports a curvature criterion that would give $g^2 > 32$ if its normalization conversion, unchecked there, stands. The Kogut–Susskind explicit threshold contrasted in §4 was withdrawn on 2026-10-02 (KS note). The bands note §1 separates the all-representation activity from the fundamental normalization, as the correction above now does.

For the Wilson action of $SU(3)$, on every spatial torus whose sides are at least 4 lattice units, the Hamiltonian $H_W = -(\hbar c/a)\log \mathcal{T}$ of the normalized transfer matrix has a gap

$$\Delta_W \;\geq\; \frac{\hbar c}{a}\, 4\log\frac{1}{176\,(e^{6/g^2}-1)} \;=\; \frac{\hbar c}{a}\Big(4\log\frac{g^2}{1056} - \frac{12}{g^2} + O(g^{-4})\Big) \qquad (g^2 \geq 1059),$$

uniformly in the volume. The threshold that Yarotsky's theorem supplies for the Kogut–Susskind Hamiltonian is $g_0^2 \sim 10^{101}$ (threshold note). The Wilson statement has an explicit threshold because the Euclidean character expansion is a polymer gas of plaquette sets covering each of their links at least twice, with no time discretization to pay for. Every polymer weight is bounded by $\rho^{|\gamma|}$ with $\rho = e^{\beta_W} - 1 = e^{6/g^2} - 1$, a bound on the sup norm of the normalized plaquette weight minus one. The Kotecký–Preiss criterion holds for $20e^2\rho \leq 0.84$, that is $\rho \leq 21/(500e^2)$, whose endpoint is approximately 5.684×10^{-3}. For this ρ the condition means $g^2 \geq 6/\log(1 + 0.84/20e^2) \simeq 1058.6$. Truncated correlations in the time direction then decay at rate at least $4\log(1/(176\rho))$ per lattice unit, because every time slab between two observables holds at least four plaquettes of one polymer. Through the transfer matrix $\mathcal{T} = e^{-aH_W/\hbar c}$, self-adjoint and positive by link reflection positivity, the rate bounds the gap.

The former headline $g^2 \geq 176$, with $\Delta_W \geq (\hbar c/a)\, 4\log(g^2/176)$, is what the same chain gives when each plaquette is charged its leading fundamental coefficient $d_f c_f/c_0 \simeq 1/g^2$. Nothing here bounds every polymer by that activity, so 176 is a leading-activity figure. The rigorous ρ exceeds the leading-order tube activity $u = c_f/c_0 \simeq 1/(3g^2)$ by the factor $2d_f^2 = 18$. That factor, the plaquette-animal entropy $20e$ and the weight $a(\gamma) = |\gamma|$ in the criterion all have room. For the same operator the Dobrushin note gives the lower rigorous threshold $g^2 > 444$; the rate here is the larger one for $g^2 \gtrsim 1.4 \times 10^3$. To leading order the gap of H_W grows like $\log g^2$ and that of H_{KS} like g^2, the two Hamiltonians being different regularizations that agree only in the continuum limit. Constants explicit; nothing promoted.

1. The polymer gas

Write the Wilson weight of a plaquette in characters,

$$f(U_p) = e^{\frac{\beta_W}{N}\operatorname{Re}\operatorname{tr} U_p} = c_0(\beta_W)\Big[1 + \sum_{r\neq 0} d_r \frac{c_r}{c_0}\chi_r(U_p)\Big], \qquad c_r = \frac{1}{d_r}\int f(U)\,\overline{\chi_r(U)}\,dU,$$

with $\beta_W = 2N/g^2$, and put $\phi(U) = f(U)/c_0 - 1$, $\phi_p = \phi(U_p)$. Expanding $\prod_p(1+\phi_p)$ over sets X of plaquettes and integrating over links, a link lying in exactly one plaquette p of X integrates ϕ_p against Haar measure and gives $\int \phi\, dU = 0$. Only sets covering each of their links at least twice contribute: closed surfaces, possibly branched (for $SU(3)$ three fundamentals can meet at a link, since $f^{\otimes 3} \ni 1$), the smallest being the six faces of a cube (§2). The polymers are the link-connected such sets, two polymers being incompatible when they share a link, and

$$Z = c_0^{|\mathcal{P}|} \sum_{\text{compatible families}} \prod_{\gamma} w(\gamma), \qquad w(\gamma) = \int \prod_{p\in\gamma} \phi_p\, dU.$$

Surfaces and the tube activity. Take a closed surface S that is a manifold (each link in exactly two of its plaquettes) and carries the fundamental with a consistent orientation. The link integrals $\int \chi_f(U_1 h)\chi_f(h^{-1}U_2)\,dh = \chi_f(U_1U_2)/d_f$ give d_f^{-1} per link, each vertex closes one index loop worth d_f, and each plaquette brings $d_f c_f/c_0$ from the expansion, so the term is

$$d_f^{V-L+|S|}\Big(\frac{c_f}{c_0}\Big)^{|S|} = d_f^{\chi(S)}\, u^{|S|}, \qquad u = \frac{c_f}{c_0},$$

with $\chi(S)$ the Euler characteristic. The dimensions enter only through $\chi(S)$, and the activity per plaquette along a tube is u. For $SU(N)$ with $N \geq 3$, the order-β_W term of the expansion below gives $d_f c_f = \beta_W/(2N) + O(\beta_W^2)$, and $c_0 = 1 + O(\beta_W^2)$, so

$$u = \frac{\beta_W}{2N^2}(1 + O(\beta_W)) = \frac{1}{Ng^2}(1 + O(g^{-2})), \qquad u = \frac{\beta_W}{18} + O(\beta_W^2) = \frac{1}{3g^2} + O(g^{-4}) \quad (SU(3)),$$

the textbook strong-coupling parameter. The fundamental coefficient $d_f c_f/c_0 = 1/g^2 + O(g^{-4})$ is d_f times larger.

All representations, rigorously. Put $x = \beta_W/N$ and expand

$$e^{x\operatorname{Re}\operatorname{tr} U} = \sum_{k\geq 0}\frac{x^k}{k!}\,2^{-k}\sum_{j=0}^{k}\binom{k}{j}(\operatorname{tr} U)^j(\overline{\operatorname{tr} U})^{k-j}.$$

The integral $\int(\operatorname{tr} U)^j(\overline{\operatorname{tr} U})^{k-j}\overline{\chi_r(U)}\,dU$ is the multiplicity $m_r(j,k-j) \geq 0$ of r in $f^{\otimes j}\otimes \bar f^{\otimes(k-j)}$, and counting dimensions, $\sum_r d_r\, m_r(j,k-j) = N^k$. Hence $c_r \geq 0$ for every r, and, every term being nonnegative,

$$\sum_r d_r^2 c_r = \sum_{k\geq 0}\frac{x^k}{k!}\,2^{-k}\sum_{j=0}^{k}\binom{k}{j}N^k = e^{Nx} = e^{\beta_W}.$$

Since $|\chi_r| \le d_r$, the character series of f converges absolutely and uniformly, to f by Peter–Weyl, and

$$\sup_U |\phi(U)| \;\le\; \sum_{r\neq 0} d_r \frac{c_r}{c_0} \sup_U |\chi_r(U)| = \sum_{r\neq 0} d_r^2 \frac{c_r}{c_0} = \frac{e^{\beta_W}}{c_0} - 1,$$

with equality at $U = \mathbb{1}$. Jensen's inequality with $\int \operatorname{Re}\operatorname{tr} U\, dU = 0$ gives $c_0 \ge 1$. Therefore every polymer weight obeys

$$|w(\gamma)| \;\le\; \int \prod_{p\in\gamma} |\phi_p|\, dU \;\le\; \rho^{|\gamma|}, \qquad \rho = e^{\beta_W} - 1 = e^{2N/g^2} - 1 \;\simeq\; \frac{6}{g^2} \quad (SU(3)).$$

This one bound covers every representation, both orientations and the branched polymers. Its leading term $2N/g^2$ is the sup norm $2d_f \cdot d_{\bar f} c_f / c_0$ of the fundamental and antifundamental terms.

2. Convergence: the Kotecký–Preiss criterion with numbers

Adjacency. In four dimensions a link lies in six plaquettes, and two distinct plaquettes share at most one link, so a plaquette shares a link with exactly $4 \times 5 = 20$ others (on $\mathbb{Z}^4$, or on a torus with sides of at least 3).

Animals. In a graph of maximal degree D, the connected sets of n vertices containing a given vertex number at most $\binom{Dn}{n-1} \le (eD)^{n-1}$. Order the neighbours of each vertex. Breadth-first search inside a set S from the given vertex, in that order, assigns to each vertex of S the subset of its D neighbour slots through which it discovers new vertices. The sequence of these n subsets in search order determines S, by replaying the search, and their sizes add to $n-1$; such sequences correspond to $(n-1)$-element subsets of the Dn slots. Then $\binom{Dn}{n-1} \le (Dn)^{n-1}/(n-1)! \le (eD)^{n-1}(\frac{n}{n-1})^{n-1} e^{-1} \le (eD)^{n-1}$, using $m! \ge e(m/e)^m$ (from $\log m! \ge \int_1^m \log t\, dt$) and $(1+\frac{1}{n-1})^{n-1} \le e$. For plaquettes $D = 20$: at most $(20e)^{n-1}$ connected sets of n plaquettes contain a given one.

Minimal size. A nonempty set X of plaquettes covering each of its links at least twice has $|X| \ge 6$. Let $p \in X$ lie in the (μ,ν) plane with corner at the origin, with sides ℓ_1 and ℓ_3 in direction μ at $x_\nu = 0$ and $x_\nu = 1$, and ℓ_2, ℓ_4 in direction ν. Since two distinct plaquettes share at most one link, the four sides are covered by four distinct plaquettes $q_i \in X \setminus \{p\}$, $\ell_i \in q_i$, so $|X| \ge 5$. Suppose $|X| = 5$. The side f_1 of q_1 opposite ℓ_1 is a μ-link displaced from ℓ_1 by $\pm e_\lambda$; it lies at $x_\nu = -1$ if $\lambda = \nu$ (the sign is fixed by $q_1 \neq p$), and at $x_\nu = 0$ with a nonzero λ-coordinate otherwise. A μ-link of q_2 or q_4 requires the plaquette to lie in the (μ,ν) plane through ℓ_2 or ℓ_4, where all μ-links have $x_\nu \in \{0,1\}$ and zero λ-coordinate; every vertex of $q_3 \neq p$ has $x_\nu \in \{1,2\}$; and the μ-links of p are ℓ_1, ℓ_3. So f_1 lies in q_1 alone, a contradiction. On a torus, sides of at least 4 keep the coordinates $-1, 0, 1, 2$ distinct.

Incompatible polymers. A polymer γ' incompatible with γ contains one of the at most $4|\gamma| \times 6 = 24|\gamma|$ plaquettes through a link of γ, so the number of such γ' of size n is at most $24|\gamma|(20e)^{n-1}$, with $n \ge 6$.

Criterion. Kotecký and Preiss (Commun. Math. Phys. 103 (1986) 491; metadata level): if $a \ge 0$ satisfies $\sum_{\gamma' \not\sim \gamma} |w(\gamma')| e^{a(\gamma')} \le a(\gamma)$ for every γ, with $\gamma' = \gamma$ included, then the cluster expansion of $\log Z$ converges absolutely and the clusters X containing a polymer incompatible with γ obey $\sum_{X \not\sim \gamma} |\Phi^T(X)| \prod_{\gamma' \in X} |w(\gamma')| \le a(\gamma)$. With $a(\gamma) = |\gamma|$ and $|w(\gamma')| \le \rho^{|\gamma'|}$ the condition follows from

$$\frac{24}{20e} \sum_{n\ge 6} \left(20e^2\rho\right)^n \;\le\; 1 \qquad \Longleftarrow \qquad x \equiv 20e^2\rho \le 0.84,$$

because $x^6/(1-x)$ is increasing on $[0,1)$ and $e > \sum_{j=0}^{5} 1/j! = 163/60$ gives, at $x = 21/25$,

$$\frac{24}{20e}\frac{x^6}{1-x} < \frac{18 \cdot 21^6}{163 \cdot 25^5} < \frac{97}{100} < 1.$$

The expansion therefore converges, uniformly in the volume, for

$$\rho \;\le\; \rho_* = \frac{21}{500e^2} \simeq 5.684 \times 10^{-3},$$

which for the rigorous $\rho = e^{6/g^2} - 1$ of $SU(3)$ reads $g^2 \ge G_{\rm KP} := 6/\log(1+\rho_*) \simeq 1058.6$.

3. Decay of correlations and the gap

Decay weights. Applying the criterion to the weights $|w(\gamma)|e^{d(\gamma)}$, whose clusters carry the same coefficients Φ^T, gives: if

$$\sum_{\gamma' \not\sim \gamma} |w(\gamma')| e^{a(\gamma')+d(\gamma')} \le a(\gamma) \text{ for every } \gamma, \quad \text{then} \quad \sum_{X \not\sim \gamma} |\Phi^T(X)| \prod_{\gamma' \in X} |w(\gamma')|\, e^{d(X)} \le a(\gamma),$$

with $d(X) = \sum_{\gamma' \in X} d(\gamma')$. With $a(\gamma) = |\gamma|$ and $d(\gamma) = \delta|\gamma|$ the count of §2 goes through whenever $xe^{\delta} \le 0.84$, that is for

$$\delta \;\le\; \delta_* = \log \frac{\rho_*}{\rho} = \log \frac{21}{500e^2\rho}.$$

Observables as decorations. Let A and B be bounded gauge-invariant functions of the spatial links of the time slices 0 and $T \ge 1$, with supports S_A, S_B, and set

$$Z(\lambda,\mu) = \int (1+\lambda A)(1+\mu B) \prod_p (1+\phi_p)\, dU, \qquad \langle A;B\rangle = \partial_\lambda \partial_\mu \log Z\big|_{\lambda=\mu=0}.$$

Expanding as in §1, A and B join the polymers that share a link with their supports. Besides the polymers γ there are decorated polymers (A,γ), (B,γ) and (AB,γ), with γ possibly empty or disconnected but $S_A \cup \gamma$ (and so on) link-connected, weights such as $\lambda \int A \prod_{p\in\gamma} \phi_p\, dU$, and incompatibility again meaning a shared link. A link of γ outside the supports still integrates to zero unless γ covers it twice, and $|\lambda \int A \prod \phi_p| \le |\lambda|\, \|A\|_\infty \rho^{|\gamma|}$. Give a decorated polymer $a = |\gamma| + |S_A|/4 + 1$ (with $|S_A| + |S_B|$ for (AB,γ)) and $d = \delta|\gamma|$. At $xe^{\delta} \le 0.84$ the undecorated polymers incompatible with an undecorated centre γ contribute at most $0.97\,|\gamma|$ to its sum, by §2, and those incompatible with a decorated centre at most $0.97\,(|S_A|/4 + |\gamma|)$, counting the $6|S_A|$ extra plaquettes through the support. The decorated polymers contribute at most the total D of $|w|e^{a+d}$ over all of them. D is finite, because their plaquette sets are unions of connected sets touching a support, counted by §2 with ratio $20e\cdot\rho e^{1+\delta} = xe^{\delta} \le 0.84 < 1$, and D is proportional to $|\lambda|$ or $|\mu|$. Take $|\lambda| \le r_A$ and $|\mu| \le r_B$ small enough, depending on A and B, that $D \le 0.18$. Undecorated polymers of nonzero weight have $|\gamma| \ge 6$, so $0.97\,|\gamma| + 0.18 \le |\gamma|$, and the criterion holds for the enlarged gas with the same δ. Every cluster that depends on both λ and μ is incompatible with the polymer $(A,\emptyset)$, so the conclusion of the criterion and Cauchy's estimate give

$$|\langle A;B\rangle| \;\le\; \frac{|S_A|/4+1}{r_A r_B}\, e^{-\delta L_{AB}},$$

with L_{AB} the least value of $\sum_{\gamma\in X} |\gamma|$ over such clusters X of polymers of nonzero weight.

Size of a connecting cluster: $L_{AB} \ge 4T$. For $0 \le t < T$ call the temporal plaquettes between times t and $t+1$ slab t. The objects of such a cluster (its plaquettes and the two supports) are connected through shared links, so they contain a chain from S_A to S_B in which consecutive objects share a link, hence a vertex. Every object other than a plaquette of slab t has all its vertices at times $\le t$ or all at times $\ge t+1$, so the chain contains a plaquette of slab t, belonging to some polymer π of the cluster. The temporal links of slab t lie outside $S_A \cup S_B$, so each one that π uses lies in at least two plaquettes of π, all in slab t. Project slab t to the spatial lattice: its plaquettes go to distinct spatial links and its temporal links to sites. The plaquettes of π in slab t become a finite graph in which every occurring site has degree at least two, which

therefore contains a cycle, and a cycle of the cubic lattice, or of a spatial torus with sides of at least 4, has at least four links. Each of the T slabs thus holds at least four plaquettes of the cluster, whatever the number of polymers in the chain and whether or not they carry A or B. With periodic time of period L_t the same argument on both arcs gives $4\min(T, L_t - T)$.

Hence $|\langle A; B\rangle| \le C_{A,B}\, e^{-4\delta_* T}$, and

$$m\,a \;\ge\; 4\delta_* \;=\; 4\log\frac{21}{500e^2\rho} \;\ge\; 4\log\frac{1}{176\rho}.$$

Transfer. The Wilson action is link-reflection positive, so the transfer matrix $\mathcal{T}$ is self-adjoint and positive (Lüscher, Commun. Math. Phys. 54 (1977) 283; Osterwalder and Seiler, Ann. Phys. 110 (1978) 440; metadata level), and $H_W = -(\hbar c/a)\log\mathcal{T}$ with $\mathcal{T}$ normalized to top eigenvalue 1 on gauge-invariant vectors. On a spatial torus that eigenvalue is simple, with eigenvector $\Omega > 0$, by Jentzsch's theorem (conditional theorem §3, item 3, which runs the same argument). For bounded gauge-invariant B on one time slice and $\psi = (B - \langle\Omega, B\Omega\rangle)\Omega$, $\langle\psi, \mathcal{T}^n\psi\rangle$ is the $L_t \to \infty$ limit of $\langle B^*; \theta_n B\rangle_{L_t}$, so it is at most $C_B e^{-4\delta_* n}$ for all n. This forces the spectral measure of ψ to vanish above $e^{-4\delta_*}$. Because $\Omega > 0$, these ψ are dense in $\Omega^\perp$, and on every spatial torus with sides of at least 4

$$\Delta_W \;=\; -\frac{\hbar c}{a}\log\big\|\mathcal{T}|_{\Omega^\perp}\big\| \;\ge\; \hbar c\, m \;\ge\; \frac{\hbar c}{a}\,4\log\frac{1}{176\rho},$$

uniformly in the volume.

Numbers for $SU(3)$.

per-plaquette activity	status	threshold	gap bound, units $\hbar c/a$
sup norm, all representations, $\rho = e^{6/g^2} - 1$	rigorous (§§1–3)	$g^2 \ge 1059$ (sufficient integer bound)	$4\log[1/(176(e^{6/g^2} - 1))]$, with leading estimate $4\log(g^2/1056)$
leading fundamental coefficient, $\rho = 1/g^2$	leading-activity figure, unproved for general polymers	$g^2 \ge 176$	$4\log(g^2/176)$

Threshold clarification, 2026-10-02. The exact sufficient KP convergence condition is $g^2 \ge G_{\mathrm{KP}}$. Its unrounded rate $4\log(\rho_*/\rho)$ is positive for $g^2 > G_{\mathrm{KP}}$ and zero at equality. The weaker displayed rate with 176 is positive only for $g^2 > G_{176} := 6/\log(177/176)$; it is zero at G_{176}. Thus 1058.6 is a rounded KP value, not the exact endpoint of the 176 rate, and 1059 is a sufficient integer bound for a positive displayed gap. This refines the earlier October 2 correction without changing its headline. Indeed $\log(1+x) > 2x/(2+x)$ for $x > 0$ (differentiate their difference); at $x = 1/176$ this gives $G_{176} < 6 \cdot 353/2 = 1059$. The number $1056 = 6 \cdot 176$ is only the linearization of that displayed condition. No equality of the rounded numbers with these exact thresholds is used. Section 2 also replaces the rounded cluster sum 0.969 by the proved upper bound 0.97; the decoration margin 0.18 in §3 still satisfies the criterion since every undecorated polymer has at least six faces. The rigorous ρ exceeds the coefficient $1/g^2$ by $2d_f = 6$: a factor 2 for the fundamental and its conjugate, and d_f from $\sup|\chi_f|$. It exceeds the tube activity $u \simeq 1/(3g^2)$ by $2d_f^2 = 18$. On an orientable fundamental surface the weight is $2d_f^\chi u^{|S|}$, with one global orientation choice, so both factors are costs of the sup norm on large surfaces. Recovering them would require bounding the link integrals of branched and multiply covered polymers by their Euler-characteristic value, which is not done here.

Room. Three constants are unoptimized: the factor 18 above; the animal entropy $20e$, where the search count is crude for plaquettes; and the weight $a(\gamma) = |\gamma|$. With $a(\gamma) = |\gamma|/10$ the

condition becomes $0.4415\,v^6/(1-v) \le 0.1$ for $v = 20e\rho e^{\delta+0.1}$, which allows $20e\rho e^{\delta}$ up to 0.59 in place of $0.84/e = 0.31$, nearly halving the thresholds in g^2. This is not carried through.

4. Relation to the Kogut–Susskind statement

H_W and H_{KS} are different regularizations of the same continuum Hamiltonian. At strong coupling their gaps differ in kind: H_{KS} has an electric term of order $g^2\hbar c/a$ and the flux loop around one plaquette costs $(8/3)g^2\hbar c/a$ for $SU(3)$; H_W has both electric and magnetic content in $-\log\mathcal{T}$, and to leading order a plaquette excitation propagates as a tube of four plaquettes per time step, with energy about $(\hbar c/a)(-4\log u) \simeq (\hbar c/a)\,4\log(3g^2)$ (a leading-order estimate). The rigorous bound of §3 has the same form with u replaced by 176ρ. The Jaffe–Witten statement asks for the continuum theory and is indifferent to the regularization, so either Hamiltonian serves for T2. The one with an explicit volume-uniform threshold is H_W: the Kogut–Susskind explicit threshold was withdrawn on 2026-10-02 (KS note), which keeps Yarotsky's existential threshold. The anisotropic limit $a_t \to 0$ that turns $\mathcal{T}$ into $e^{-a_t H_{\mathrm{KS}}/\hbar c}$ sends the temporal plaquette activity to one, outside the expansion, which is why the Kogut–Susskind threshold needs a genuinely Hamiltonian expansion and why Yarotsky's time discretization pays what it pays.

5. Consequence for STATE

The rigorous strong-coupling edge of this route is $g^2 \ge 1059$: for the Wilson transfer matrix and $SU(3)$ the theory is gapped uniformly in the volume (spatial tori with sides of at least 4), with $\Delta_W \ge (\hbar c/a)\,4\log[1/(176(e^{6/g^2}-1))]$. The figure 176 is the leading-activity reading only. For the same operator the Dobrushin note's $g^2 > 444$ is the lower rigorous edge, and this note's rate is the larger one for $g^2 \gtrsim 1.4\times 10^3$. The unoptimized constants are the sup-norm factor 18, the animal entropy $20e$ and the weight $a(\gamma)$. On the weak side no step is proved with explicit constants; the necessary window of the operator-inequality note is nonempty only for $1/g^2 \gtrsim 10^2$ (crude constants). The intermediate region, from $g^2 \sim 10^{-2}$ to the rigorous strong edge (444 by Dobrushin, 1059 here), is where the gap forms. Those integer figures are sufficient bounds: the exact KP condition and the stricter condition for positivity of the rounded 176 rate are distinguished in §3. LLM.md and the position note already carry the October 2 headline correction; their leading estimates must retain their approximation signs.